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Buffering and H^+ ion dynamics in muscle tissues

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Abstract

After anaerobic activity with release of large quantities of intermediary metabolic end products, further energy production critically depends on rapid elimination of H^+ ions from the muscle tissues to secure key enzymatic activities. The involved processes, interactions and interrelationships of mechanisms have been analyzed on the basis of a physiological model and available experimental data. The H^+ elimination from muscle tissue is a multifactorial process primarily governed by the capillary H^+ transport capacitance, effected by buffering capabilities of intracellular and capillary fluids compartments, by dynamically interrelated regulation of intracellular and extracellular pH, by the magnitude and quality of convective perfusional transfer and further factors. Model calculations strongly resemble experimental data obtained in isolated perfused muscles and suggest that discrepancies between disparate literature data are attributable to experimental limitations.

Keywords: Acid–base; Regulation; Skeletal muscle; Buffer; Capacity; Muscle; Skeletal; Buffering; H^+ release; Proton; Transport capacitance; Elimination

1. Introduction

Generally, three main mechanisms are responsible for acid–base adjustment in vertebrate organisms: buffering, transmembrane and transepithelial ion transfer processes, and the adjustment of P_{CO_2}. Buffering of surplus H^+ ions is a valuable mechanism for minimizing transient acid–base disturbances, while acid–base relevant ion transfer is required to finally void the metabolic dissipation of H^+ from the body fluids, either by promoting metabolic processing in other compartments or by transepithelial elimination. Adjustment of ventilation, of irrigation and/or perfusion of gas exchange structures serves the adjustment of a generally constant P_{CO_2}, independent of the production rate of CO_2 as the volatile end product of aerobic metabolism. The effects of these processes have been elucidated by numerous studies under various conditions for the bulk extracellular fluid (for review e.g. Heisler, 1986c, 1990a, 1993). Their dynamic interplay within the systemic tissues, however, is much less understood, largely due to the complex interaction and interdependence of mechanisms in the chain of H^+ elimination from cells via interstitial space to bulk extracellular fluid, which are inaccessible to present methodology. This paper is

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focused on analyzing a few aspects of such interactions and mutual limitations on the basis of a physiological model.

2. Buffering in tissues

Since definition of the 'buffer value' as the amount of H^+ required to titrate system pH by 1 unit (Van Slyke, 1922), numerous data have been presented for blood and plasma, but also for the intracellular compartments of various tissues (for review: Heisler, 1986a). Distinction has to be made between the chemical buffer value in systems closed for all involved components and the buffer slopes determined by various approaches in physiological systems open for acid–base relevant ions and for constituents of the buffer systems, as well as between the buffering capabilities of non-bicarbonate buffers (NB) and of the bicarbonate system (cf. Heisler, 1986a). The CO_2/bicarbonate buffer system will accept H^+ ions related to non-volatile anions, but is, in contrast to NB-buffers, incapable of handling H^+ as a dissociation product of CO_2, originating from its own chemical equilibrium.

The non-bicarbonate buffer values of intracellular body compartments are generally much higher than those of interstitial fluid and plasma, due to relatively high protein and phosphate concentrations (cf. Heisler, 1986a,d). Although the plasma protein concentration is relatively high, these proteins contain relatively fewer amino acid groups with pK values in the physiological pH range than haemoglobin, myoglobin and other intracellular proteins. High intracellular buffer values serve steady-state elimination of CO_2 and H^+ with small gradients to the bulk extracellular fluids.

In extracellular bulk fluids, the highly concentrated bicarbonate is a valuable buffer for non-volatile H^+ ions, as long as the resulting CO_2 can be eliminated during passage of the gas exchange organ (cf. Heisler, 1986a). In tissues, the role of bicarbonate is fairly small, mainly due to the much lower initial concentration (1/2–1/4), but even more so because of the virtually closed buffer system conditions. Since CO_2 cannot be directly voided from intracellular (i), interstitial (int) and capillary compartments (c), bicarbonate buffering of even small amounts of H^+ will raise P_{CO_2} in tissues to extreme values. Similarly, any further support of acid–base regulation by active, compensatory adjustment of P_{CO_2} (cf. Heisler, 1986a) is limited to extracellular bulk fluids passing gas exchange structures, but is made impossible in the intracellular compartments and in blood during capillary transit.

3. Transmembrane transfer of acid–base relevant ions

Typically high non-bicarbonate buffering capabilities of systemic tissues (cf. Heisler, 1986a,d) are most suitable to ameliorate the effects of sudden H^+ release in the tissues within a few milliseconds, but will quickly be exhausted if not relieved by elimination of H^+-equivalents from the site by other mechanisms. Particularly with stress-induced production of acid–base relevant ions in large amounts, significant excursions of pH from set point values cannot be avoided, such as after changes in temperature (e.g. Heisler and Neumann, 1980; Heisler et al., 1980), during environmental hypercapnia (e.g. Heisler and Piiper, 1972; Heisler, 1986b) and particularly after exhausting muscular activity (Holeton and Heisler, 1983; Holeton et al., 1983; Benadé and Heisler, 1978). The induced shifts in pHi may well significantly reduce or even completely inhibit further energy production by the lack of key enzymatic activity (e.g. of PFK; cf. Heisler, 1990b).

Accordingly, elimination of H^+ from the site of energy production is the only option for steady-state maintenance of undisturbed tissue function. Transfer of acid–base relevant ions across cell membranes has first been directly demonstrated during hypercapnia in muscles tissues (Heisler and Piiper, 1972), but also occurs during lactacidosis (Benadé and Heisler, 1978) and other acid–base perturbations of intracellular compartments (for review: Heisler, 1986a,b).

4. H^+ elimination after anaerobic muscular activity

Efficient elimination of H^+ is suitable to maintain cellular energy production even when H^+ ions are produced at high rates during anaerobic muscular energy production. Transfer across cell membranes can apparently be performed at very high rates, as established in thin muscle preparations suspended in Ringer solution

(Benadé and Heisler, 1978). Due to the experimental obligation to completely satisfy oxygen demand by diffusion, thin muscle preparations facilitate minimal interstitial diffusion pathways and thus small diffusional resistances for elimination of acid–base relevant ions into the suspending bulk Ringer fluid. Accordingly, the measured net intracellular to extracellular H^+ transfer rates have to be considered as mainly, if not exclusively limited by the membrane transfer process itself. After bursts of anaerobic activity with intracellular accumulation of lactic acid, dissociated H^+ ions are extruded with an average half time of 39 or 25 s (rat diaphragm or frog sartorius muscle, respectively; Benadé and Heisler, 1978). Lactate, on the other hand is dissipated from intracellular compartments with half times of 9 and 22 min, respectively, evidently by a mechanism different from that for the H^+ ions.

In comparison with the very high transfer rates obtained in unlimited Ringer-suspended preparations (see above), the occurrence of H^+ ions in the extracellular space after intense activity is often much delayed in intact animals. During and after whole animal muscular activity, the elimination kinetics of H^+ are frequently found much closer to that of lactate (e.g. Hirche et al., 1975; Mainwood et al., 1972; Mainwood and Worsley-Brown, 1975) than to the H^+ flux rates in isolated, suspended muscle preparations. However, the amount of surplus H^+ ions observed in the circulating blood is generally not equivalent to the amount of lactate in those experiments.

Assimilation of transfer kinetics for H^+ and lactate in whole animal organisms is not too surprising when quasi-steady state conditions for the elimination processes are attained. In fish, reduction of bulk extracellular pH by elimination of a relatively small fraction of the H^+ produced in muscle tissues leads to a new equilibrium between intracellular and extracellular pH, limiting any further H^+ transmembrane transfer to the rate of excess H^+ removal from the organism ('equilibrium limitation', cf. Holeton and Heisler, 1983). This removal is restricted to the H^+ equivalent transepithelial transfer at the gills (for review: Heisler, 1986b). In generally air breathing terrestrial animals the limit is essentially the rate of further aerobic lactic acid processing by either glycogen resynthesis or oxidation to CO_2. Both pathways process stoichiometric amounts of H^+ and lactate, thus leading towards similar elimination kinetics from muscle tissues by definition. Since lactic acid is produced in whole animal organisms with a relatively retarded time course rather than in a burst mode, specific evaluation of the transmembrane H^+ transfer rate is made difficult or even impossible by quasi-steady state conditions.

5. Transfer dynamics in isolated perfused muscle preparations

In order to avoid quasi steady-state conditions and to achieve a more or less rectangular onset of the vascular efflux process of H^+ and lactate from the intracellular compartment, experiments were performed utilizing preparations of blood-perfused dog gastrocnemius muscle. Surgically isolating the vascular arterial supply and venous efflux from the muscle allowed clear separation of lactic acid production and accumulation during circulatory arrest from the subsequent elimination of H^+ and lactate during reperfusion. Although a small fraction of excess H^+ ions may have already been transferred to the small interstitial space during circulatory arrest, the majority will have been located in the intracellular space at the beginning of the efflux period. Analyzing the amounts of H^+ and lactate eliminated in the muscle effluent blood showed a wide variety of H^+ kinetics among individual experiments, while the elimination characteristics of lactate appeared fairly uniform (N. Heisler, H. Weitz and A.J.S. Benadé, unpublished data). This pattern resulted in ratios of efflux half times of lactate/H^+ ($t_{0.5\,\mathrm{lact}}/t_{0.5\,\mathrm{H}^+}$) ranging from 0.8 to 8. Closer analysis of tissue perfusion rate during the efflux period indicated that the elimination of H^+ was somewhat correlated with blood flow through the tissue capillaries. In contrast, lactate elimination was independent of perfusion within the studied range, suggesting that there is no perfusion limitation for the rate at which lactate is transferred across the cell membrane.

The correlation between H^+ efflux from the muscle tissue and perfusion rate strongly suggests that H^+ translocation across the muscle cell membrane may occur rapidly, as found in Ringer-suspended preparations, but is limited by the rate of perfusion. However, perfusion of the capillary bed was not the only factor involved. The rate of H^+ elimination from the tissue was influenced as well by quite a number of additional factors, such as blood-Hct, acid–base parameters of inflowing perfusate (P_{CO_2}, [HCO_3^-], plasma pH), time elapsed after intracellular H^+ production, and the profile of perfusate flow as a function of time after anaerobic activity. The general pattern suggested the overall H^+ capacitance of the perfusate as the link between intracellular compartments and bulk extracellular fluids to be the superior parameter in limiting the kinetics of H^+ elimination from the tissues.

6. H^+ capacitance of the perfusate

For a monocompartmental system, the H^+ capacity is the product of buffer value, the change in pH applied during H^+ buffering and the volume of the system (cf. Heisler, 1986a). During capillary transit in muscle tissues, the conditions are more complex. As long as membrane transfer is not limiting the process, the extracellular fluid takes up H^+ ions translocated across the cell membrane until pH is reduced to attain a new equilibrium between intracellular and extracellular pH, by definition with no further transfer of substance regardless of the nature of involved transfer mechanism(s). The amount of H^+ required to attain this new pH equilibrium is then a function of the total buffer value of the available extracellular space, adding up from the contributions of individual non-bicarbonate buffers and of bicarbonate.

While the contribution of the non-bicarbonate buffers in the perfusate (Hb, phosphates, plasma proteins) is relatively predictable, the amount of H^+ buffering by bicarbonate in the virtually closed system of capillaries plus interstitial and intracellular fluid compartments depends largely on compartmental P_{CO_2}'s and total CO_2 dynamics among the involved compartments (cf. Heisler, 1986a). According to its high diffusivity, CO_2 equilibrium is readily achieved during capillary transit so that perfusate P_{CO_2} is a direct function of the CO_2 level in the tissue and vice versa. This in turn is related to the amount of H^+ converting HCO_3^- into CO_2 in the respective compartments, as well as to the fraction of H^+ already removed from the ICS and to the aerobic tissue CO_2 production rate.

According to the limited contribution of bicarbonate during closed system transit of the capillaries, non-bicarbonate (NB) buffering is expected to take up the majority of H^+ ions released from the ICS to the perfusate. Non-bicarbonate buffering is a function of the concentration of haemoglobin ([Hb]) as the main carrier of buffer groups with p*K*'s in the physiological pH range. At least two mechanisms may affect capillary [Hb] during transit: the flux of water following osmotic gradients established by intracellular accumulation of additional osmolytes (lactate$^-$), and plasma skimming, shunting of a variable fraction of well buffered erythrocytes past the ICS–perfusate interface. Water shifts into the ICS will elevate the NB-buffer value of the perfusate, thus supporting elimination of H^+, but plasma skimming will reduce NB-buffering with the effect of reduced H^+ capacitance. Both of these mechanisms are difficult to quantify in a perfused muscle preparation or in vivo.

In addition to these factors of physico-chemical properties, the capacity for H^+ ion removal from the tissue is affected to a large extent by the volume of available extracellular fluid. As long as complete equilibrium can be achieved during transit time, the elimination capacitance for H^+ (per unit of time) is a function of tissue perfusion rate. The overall capacitance of the perfusion module of H^+ removal is accordingly resembled mainly by the two factors of [Hb] with its potent NB-buffering and of the rate of capillary perfusion.

7. Model of H^+ elimination from muscle tissues

Evidently, a number of interrelated factors contribute to the rate of H^+ elimination from the muscle cells, with the individual influence depending on the prevalent conditions. Experimental analysis of isolated perfused muscle preparations indicates an important role of perfusion rate, but other factors are involved. Accordingly, the facettes of tissue H^+ elimination have been modelled on the basis of data available for specific aspects and contributing mechanisms. The amounts of H^+ translocated per unit of time along the pathway from muscle cells via membrane, interstitial space, capillary compartment to the bulk extracellular fluid pool was analyzed utilizing relevant mathematical relationships established for distribution equilibria and buffering processes (e.g. Heisler, 1978, 1984, 1986a; Benadé and Heisler, 1978; Heisler and Neumann, 1980). Since these relationships contain linear as well as exponential/logarithmic terms, closed mathematical solution remained impossible and the model was solved by iterative processes for specified time and stoichiometric intervals. Experimental data for the following parameters and variables were utilized as a basis for the modelling process:

- buffer values for mammalian skeletal muscle intracellular space and interstitial fluid (Heisler and Piiper, 1971; Heisler, 1986a)
- buffering and equilibrium properties for mammalian plasma and erythrocytes (Heisler and Schorer, 1970; Nissen and Heisler, 1973)
- H^+ transmembrane transfer characteristics of mammalian muscle cells as a function of various parameters (based on Benadé and Heisler, 1978, and unpublished data)
- pH_i/pH_e equilibrium conditions in mammalian muscle tissues (Heisler, 1975; Gonzalez, 1986)
- perfusion as a function of time after anaerobic exercise in isolated perfused mammalian muscle (N. Heisler, H. Weitz and A.J.S. Benadé, unpublished data; N. Heisler, H. Weitz and K. Wiese, unpublished data)
- fractional compartment volumes of mammalian muscle tissues (Heisler, 1975; N. Heisler and A.J.S. Benadé, unpublished data; K. Wiese, H. Weitz and N. Heisler, unpublished data)
- Constants for $CO_2/[HCO_3^-]$–Henderson–Hasselbalch equilibria (α_{CO_2} and pK_1') (Heisler, 1986a, 1989) for appropriate parameters (e.g. ionic strength, molarity, temperature, [Pr], $[Na^+]$).

The model is characterized by five interrelated fluid compartments, intracellular space (ICS), interstitium (INT), capillary plasma (cPL) and erythrocytes (cERY), and the bulk extracellular fluid (ECS). Volumes of muscular compartments, ICS and INT, are treated as constant, whereas the available volumes of the moving compartments, cPL and cERY, depend on the rate of perfusion. For this version of the model, volume changes by water movements following osmotic gradients as well as changes in tissue temperatures during the efflux process and of perfusate temperature during capillary transit were not taken into consideration. Also, this version does not attend to heterogeneities in local tissue perfusion (e.g. Marconi et al., 1984, 1988; Piiper et al., 1985, 1989; Ceretelli et al., 1984) and in the amount of excess H^+ ions in the ICS (Heisler, 1988). The non-bicarbonate buffering capacity of the ICS was treated constant, the buffering abilities of the other compartments were treated variable according to the quality of perfusate with respect to haematocrit or [Hb], plasma proteins, interstitial proteins and other non-bicarbonate buffers, as well as the bicarbonate concentration for the respective conditions. The model incorporates all movements of H^+ (or other acid–base relevant ions, such as HCO_3^- or OH^-) and total CO_2 among the studied compartments, but does not take into account any transfer of other substances, such as constituents of NB-buffer systems.

8. CO_2 equilibria

The model indicates considerable acidification of mammalian muscle ICS upon anaerobic activity. Production of 40 mmol/(kg muscle tissue) lactic acid for example results in a shift of intracellular pH from 6.9 to about 6.3. Before any transfer of H^+ or CO_2 to the perfusate takes place (i.e. total CO_2, $C_{CO_2} = [HCO_3^-] + \alpha_{CO_2} \times P_{CO_2}$, is still the same as before the activity), P_{CO_2} in ICS and INT rises to values of 115 mm Hg (15.3 kPa) and more due to conversion of HCO_3^- into molecular CO_2 (Fig. 1, upper panel). With reperfusion, the perfusate is acidified to an equilibrium pH_{pl} of about 6.5 by H^+ transfer. On the basis of constant C_{CO_2}, $P_{CO_{2pl}}$ is elevated to values of more than 220 mm Hg (29.3 kPa), but according to diffusion of CO_2 into the muscle tissue the gradient tapers off to attain an intermediate value closer to the original muscle P_{CO_2} (Fig. 1). With supply of fresh perfusate and buffering of H^+ there is a continuing flux of CO_2 into the muscle, which is only slowly reversed during later stages of the H^+ elimination process (ΔC_{iCO_2}, Fig. 1). The extent of maximal P_{CO_2} rise is related to the fraction of bicarbonate/non-bicarbonate buffering and thus P_{CO_2} rises to peak values with NB-free bicarbonate Ringer as a perfusate.

9. Effect of rate and quality of muscle perfusion

The efflux of H^+ ions from the muscle tissue into the ECS is largely depending on the rate of perfusion. For a constant flow rate of 1.5 l min^{-1} kg^{-1}, the highest flow rate observed to isolated dog gastrocnemius after extensive anaerobic activity in our laboratory, the model indicates extremely fast H^+ efflux kinetics (Fig. 2) with

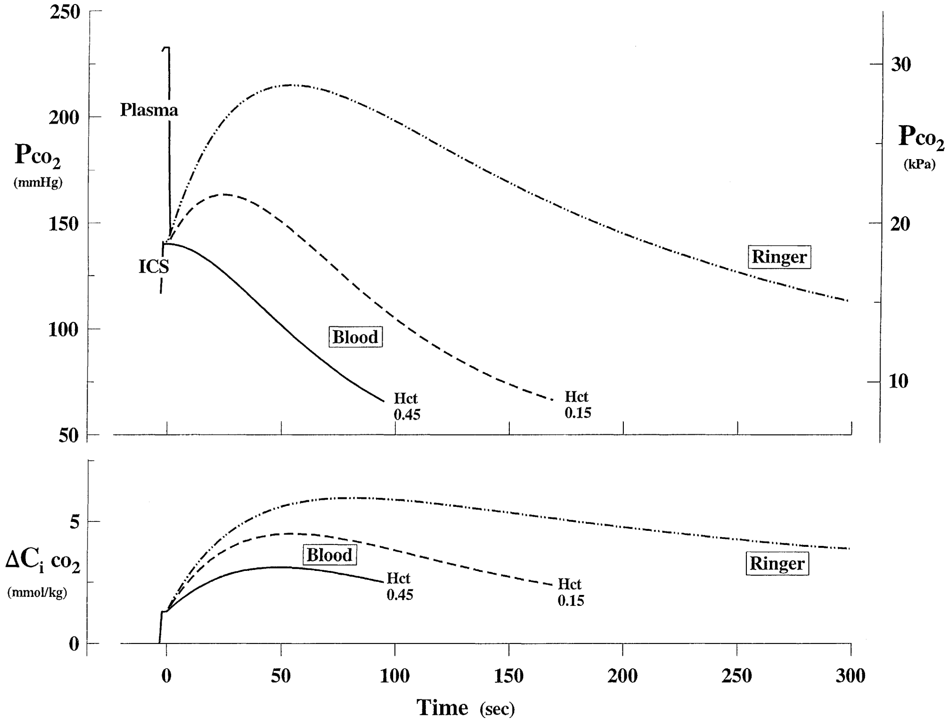

Fig. 1. Capillary P_{CO_2} and changes in intracellular total CO_2 (C_{iCO_2}) during the H^+ elimination process in muscle tissue. Perfusion with blood (Hct 0.45 or 0.15, respectively) and bicarbonate Ringer solution. Model calculations based on experimental data (see text).

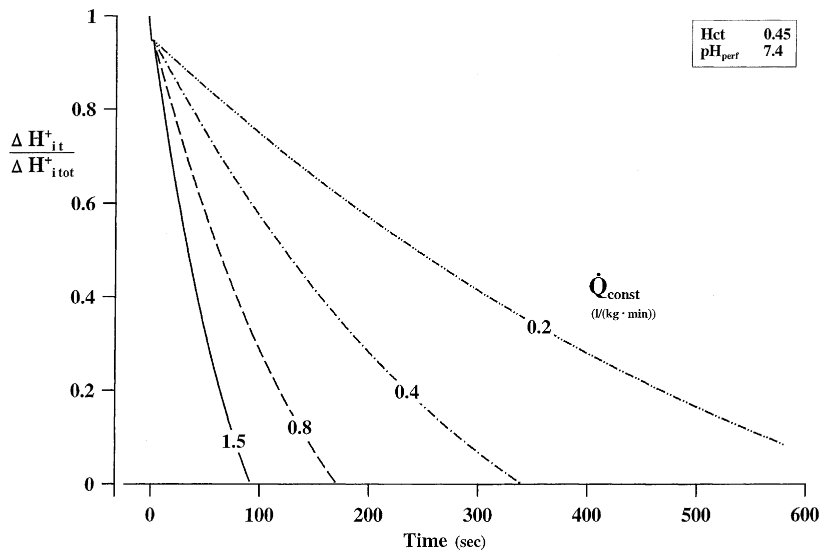

Fig. 2. Efflux kinetics of H^+ from muscle cells. Fractional amount of intracellular ('*i*') H^+ ($\Delta H^+_{i\,t}/\Delta H^+_{i\,tot}$) over total amount ('tot') as a function of time. Constant blood perfusion (Hct 0.45) at indicated constant rates ($\dot{Q}_{const}$). Model calculations (see text).

half times in the range of 30 s, comparable to those obtained in isolated Ringer-suspended muscles (Benadé and Heisler, 1978). At lower rates of constant muscle perfusion, H^+ elimination is predicted considerably retarded due to the limitation of H^+ capillary transport capacitance, with efflux half times of more than 240 s at a flow rate of 0.2 l min^{-1}. This limitation by the effective H^+ capacitance is highly escalated by anaemic perfusate ([Hb] = 5 g%, Hct 0.15), and even further by perfusion with plasma (Fig. 3). The shifts of the efflux curves are not proportional to the change in [Hb], indicating that basic levels of perfusate non-bicarbonate buffering are sufficient to still maintain a relatively effective removal of H^+ ions. The essential role of non-bicarbonate buffering for the elimination process is well illustrated by extremely delayed efflux kinetics caused by perfusion with NB-buffer-free bicarbonate Ringer solution, a type of perfusate that has often been used for H^+ efflux experiments on isolated perfused muscle preparations (Fig. 3). Although plasma NB-buffering has to be considered rather poor, even duplication of Ringer flow rate cannot boost H^+ elimination to the level obtained by plasma perfusion (cf. Fig. 3; $\dot{Q}_{const}$, curves for constant flow rate of plasma 0.8 and Ringer 1.5 l/(kg min)). Use of bicarbonate Ringer also affects the characteristics of the efflux process shown by a flat start in the efflux curve as compared to NB-buffered perfusates before similar kinetics are attained in the later efflux process (cf. curves for plasma 0.8 and Ringer 1.5 l min^{-1}).

The elimination kinetics are also affected by the acid–base composition of the perfusate. At lower than normal pH of 7.2, H^+ elimination is retarded as compared to pH 7.4, conditions that are characteristic for extended whole body exercise with activity of a major proportion of musculature (Fig. 4). Also changes in inflowing perfusate [HCO_3^-] at constant pH (compensated by P_{CO_2}) provide similar though smaller effects, with lower concentrations (higher P_{CO_2}) impeding the efflux process (Fig. 4). The differential sensitivity to changes in pH and [HCO_3^-] clearly indicates a much higher contribution of NB-buffers as opposed to HCO_3^- for the H^+ elimination process (Fig. 4).

10. Effect of intracellular buffering

Intracellular NB-buffering is clearly an important factor in muscle tissues in order to limit extensive pH shifts expected upon massive lactic acid production in

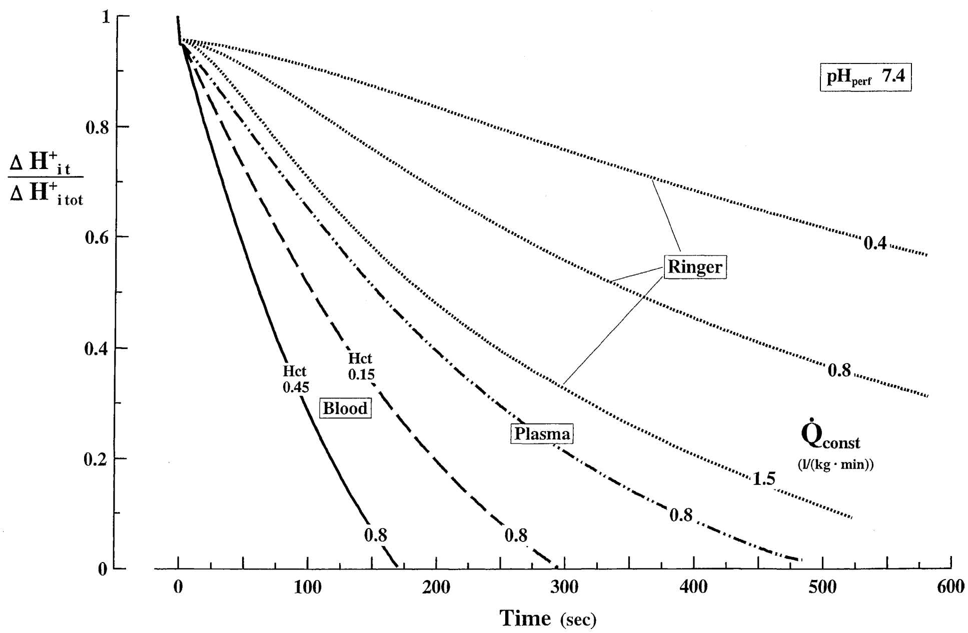

Fig. 3. Effect of perfusate quality and flow on kinetics of H^+ elimination from muscle tissues. Fractional amount of intracellular H^+ ($\Delta H^+_{i\,t}/\Delta H^+_{i\,tot}$) as a function of time. Perfusion with blood (Hct 0.45 or 0.15), plasma and bicarbonate Ringer solution. Model calculations (see text).

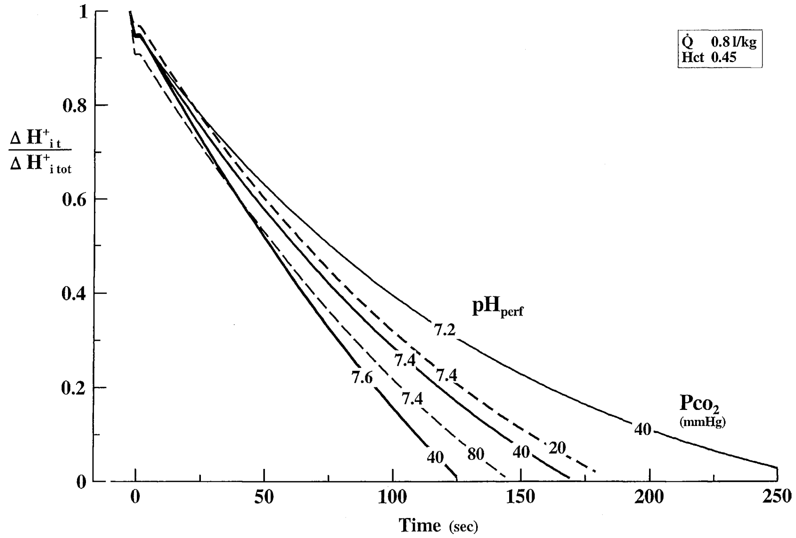

Fig. 4. Effect of perfusate pH on kinetics of H^+ elimination from muscle tissues (solid lines). Fractional amount of intracellular H^+ ($\Delta H^+_{i\,t}/\Delta H^+_{i\,tot}$) as a function of time. Effect of changes in [HCO_3^-] are indicated by changes in P_{CO_2} at constant pH (dashed lines). Model calculations (see text).

the cells. On the other hand, high intracellular buffer values affect the distribution of H^+ ions between intracellular and extracellular compartments, which in turn will indirectly influence the amount of H^+ taken up and carried away into the bulk fluid by the perfusate. With all other variables kept constant, the efflux kinetics are inversely related to the intracellular buffer value, i.e. a larger intracellular $ß_{NB}$ will retard the H^+ ion efflux process (Fig. 5). As discussed above, this effect is opposite to that for changes in extracellular NB-buffering (blood versus plasma or bicarbonate Ringer) and is also not proportional to the absolute buffer value.

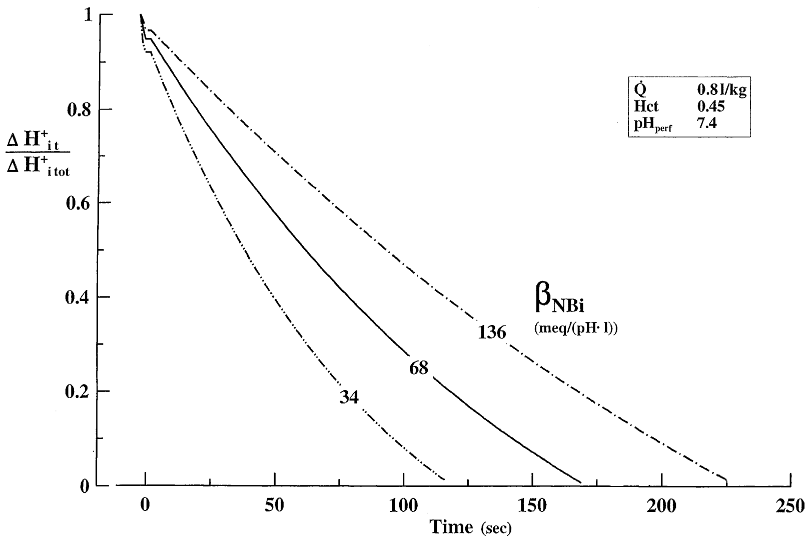

Fig. 5. Effect of intracellular NB-buffer value. Fractional amount of intracellular H^+ ($\Delta H^+_{i\,t}/\Delta H^+_{i\,tot}$) as a function of time. Model calculations (see text).

For a given preparation, the concentration of intracellular non-bicarbonate buffers is short-time constant, but relatively large differences in intracellular NB-buffer values are observed among species. On the basis of the present model, the generally lower values found in burst-activity species, such as fishes (35–50 versus 65–78 mequiv./(l· pH) for mammals; cf. Heisler, 1986d) are advantageous in terms of H^+ elimination from the muscle tissues. This may serve mainly anaerobic predatorial or flight burst activities in animals with low capillary density and accordingly low oxidative metabolic capacity, resulting in quick recovery for further muscular activity. On the other hand, a lower intracellular buffer value may limit the amount of lactic acid production and accordingly the anaerobic energy production by pH-induced inhibition of key enzymes like PFK (cf. Heisler, 1990b).

11. Membrane versus perfusion/equilibrium limitation

The model clearly indicates that the elimination of H^+ is largely affected by quantity and quality of perfusion, which may be summarized as the perfusion/H^+ capacitance module of the efflux process. The membrane transfer as the second potentially limiting module is a rapid process, as indicated by the data obtained in isolated, suspended muscle preparations (Benadé and Heisler, 1978). The elimination kinetics obtained from such preparations can be considered to almost exclusively reflect the transfer rate of the membrane module, as long as the interstitial diffusion distances are small and transferred H^+ are removed from the surface interface of the muscle by adequate irrigation.

Introducing the membrane transfer module into the model calculations shows that the membrane transfer rate is generally not a limiting factor in excess to the limitations imposed by perfusion rate and perfusate buffering characteristics. Only at very high rates of constant perfusion with high perfusate buffer capacity the membrane transfer becomes limiting during the lower end of the efflux kinetics (Fig. 6, broken lines), related to the progressive reduction in membrane transfer rate as a function of the amount of H^+ already eliminated from the ICS (Benadé and Heisler, 1978). This membrane limitation is particularly apparent due to the constantly high H^+ capacitance at constant perfusion, a condition often applied in relevant studies. In physiological conditions, however, perfusion is not regulated

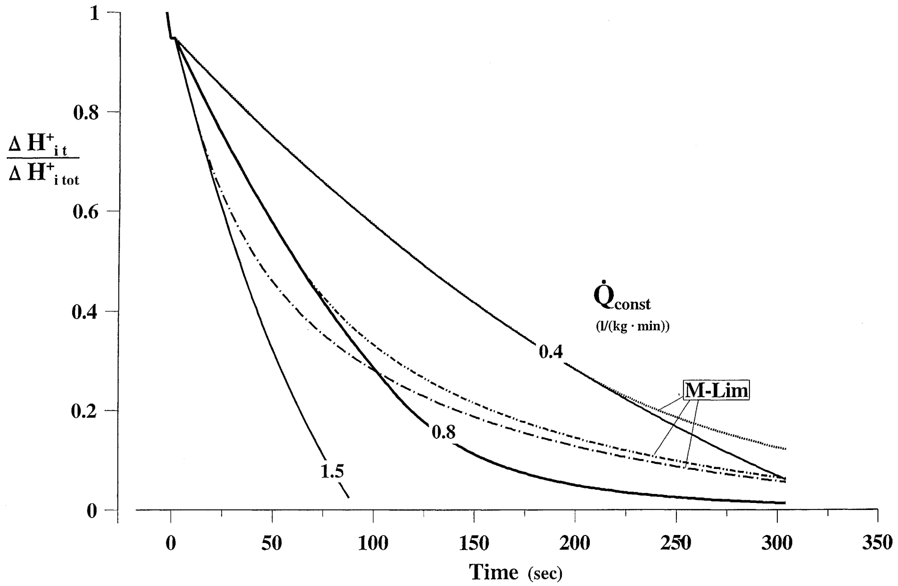

Fig. 6. Efflux kinetics of H^+ from muscle cells. Blood perfusion (Hct 0.45) at indicated constant rates ($\dot{Q}_{const}$). Solid lines represent efflux limits provided by the H^+ transport capacitance. Dash-dotted, dash-double dotted and pointed lines characterize membrane limitation (M-Lim). Model calculations (see text).

constant, but modulated to meet tissue demands. Accordingly, the model was adjusted to variable perfusion by input of data on the perfusion/time profile determined in dog gastrocnemius muscle. After reestablishment of perfusion the flow rate steeply rises to a maximum and then tapers off in a multi-exponential curve (Fig. 7, lower panel). These flow characteristics considerably affect the H^+ elimination from the muscle tissues. With physiological flow/time profiles with peak values of 1.5 and 1.0 l/(kg min), the perfusion limitation curves track the membrane limitation curves, suggesting that perfusion under autoregulatory conditions is at least partially adjusted to the rate at which H^+ can be transferred across the cell membranes (Fig. 7, upper panel). With these flow/time profiles (Fig. 7, lower panel), the limitation by the membrane transfer vanishes below about 1.0 l peak flow rate, when limitation by the H^+ capacitance module takes over (Fig. 7).

12. Physiological significance

The pattern projected by this model with its wide variability of the H^+ ion elimination kinetics is reflected in the data of numerous experimental studies. Reports on H^+ tissue efflux in the literature are quite heterogeneous, with ratios between H^+ and lactate elimination in isolated muscle preparations varying between 40 and values slightly below 1 (e.g. Hirche et al., 1971, 1975; Mainwood et al., 1972; Mainwood and Worsley-Brown, 1975; Benadé and Heisler, 1978; Holeton and Heisler, 1983; Holeton et al., 1983; Piiper et al., 1972; Wardle, 1978; Boutilier et al.,

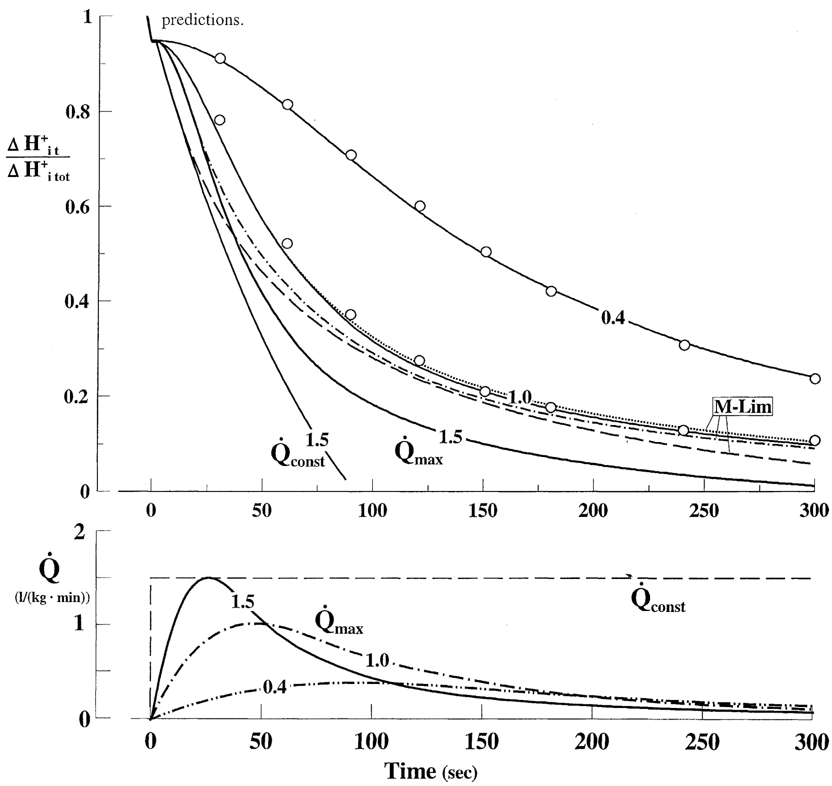

Fig. 7. Efflux kinetics of H^+ from muscle cells. Blood perfusion (Hct 0.45) at indicated constant or variable rates according to lower panel flow/time profiles (characterized by maximal flow rate, $\dot{Q}_{max}$; experimental data) ($\dot{Q}$, l/(kg min)). Solid lines represent efflux limited by H^+ transport capacitance. Dashed, dash-dotted and pointed lines characterize membrane limitation (M-Lim). Model calculations (see text); Open circles represent experimental data from dog gastrocnemius muscle (perfusate flow rates 1.01 and 0.43 l/(kg min) of blood with Hct of 0.45) for comparison with model predictions.

1985). In whole animal experiments the results are spread out to even a much wider range (for ref. Benadé and Heisler, 1978).

Many of these disparate literature data can be incorporated into the general pattern provided by the present model as a unifying link, as long as relevant data on key parameters are available (for a first approximation data on perfusion rate and quality of perfusate). As described above, the H^+ elimination kinetics in isolated perfused dog gastrocnemius muscle referred to the lactate elimination may differ by a factor of ten. In spite of such huge differences, theoretical analysis on the basis of the experimental boundary conditions is capable to closely characterize the inherent type of limitation (Fig. 7, two sets of efflux data). Often, closer analysis is not required to evidence that limitations imposed by too adverse experimental conditions have prevented collection of physiologically relevant data, such as during low-rate perfusion of isolated muscles with NB-free Ringer solution.

The present model indicates that the elimination of H^+ ions from muscle tissues is a multifactorial process, largely depending on quality and quantity of capillary perfusion. Transfer of H^+ out of the tissue is mainly a function of the H^+ transport capacitance of the available extracellular space as a complex substrate of numerous factors like hydrodynamic capillary flow, buffering of the involved fluid compartments, intercompartmental CO_2 flow, H^+ distribution equilibria and various other contributing factors. Establishment of this model has also evidenced that determination of the membrane transfer capacity requires extensive experimental efforts and consideration in terms of providing an adequate (i.e. a larger than the membrane transfer rate) H^+ transport capacity between membrane interface and bulk extracellular fluid in order to arrive at the actual membrane-located limitations of H^+ transport.

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